L1.6 Trig Derivatives & the Chain Rule

§2.4–2.5 — what you need to know

The six trig derivatives

LEARN THESE THREE	**GENERATE THESE THREE**
$\dfrac{d}{dx}\big(\sin x\big) = \cos x$	$\dfrac{d}{dx}\big(\cos x\big) = -\sin x$
$\dfrac{d}{dx}\big(\tan x\big) = \sec^2 x$	$\dfrac{d}{dx}\big(\cot x\big) = -\csc^2 x$
$\dfrac{d}{dx}\big(\sec x\big) = \sec x \tan x$	$\dfrac{d}{dx}\big(\csc x\big) = -\csc x \cot x$

"the derivative of a co-function is always negative"

Take the row on the left, put *co-* on every function, attach a minus sign. The co-function derivative is the co-function **of the derivative**: $\sec x \tan x \to \csc x \cot x$, negated.

RADIANS AND DEGREES

$\dfrac{dy}{dx}$ is an output change over an input change, so differentiating a trig function puts an **angular** measure and a **linear** one in the same ratio.

Linear and angular measures mixing forces radians. Where nothing mixes, the choice is free. Differentiating is itself the mixing — so every formula above is in radians.

The receipt: $\lim_{\theta\to 0} \frac{\sin\theta}{\theta} = 1$ in radians but $\frac{\pi}{180}$ in degrees — which is why $\frac{d}{dx}\sin(x^\circ) = \frac{\pi}{180}\cos(x^\circ)$. Reading a graph or naming $\sin 30^\circ$ mixes nothing, so there you are free.

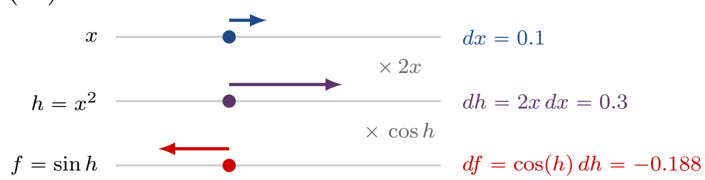
DIFFERENTIALS

dx, dy and $d\big(f(x)\big)$ **live on their own.** dx is a nudge in the input, dy is the nudge that comes back out, and $\dfrac{dy}{dx}$ is a *ratio of two of them* — not one symbol.

$\dfrac{dy}{dx} = f'(x) \implies dy = f'(x)\,dx.$ At $x = 1$ with $dx = 0.1$: $y = x + 1$ gives $dy = dx = 0.1$, the nudge straight through; $y = x^2$ gives $dy = 2x\,dx = 0.2$ — **the same nudge, doubled on the way out.**

THE CHAIN RULE

Nudge the input and watch the nudge come out the far end **changed by a factor at each stage**. For $y = \sin\left(x^2\right)$ at $x = 1.5$ with $dx = 0.1$:



Substitute the middle line into the bottom one — $df = \cos\left(x^2\right) \cdot 2x\,dx$ — and divide by dx:

$$\boxed{\frac{df}{dx} = f'\big(h(x)\big) \cdot h'(x)}$$

f' is **evaluated at the inside**, times the derivative of the inside. **A negative factor sends the nudge back out the other way**, as above.

WHICH RULE RUNS FIRST

The outermost operation names the first rule. Read what the expression *is* before differentiating any piece of it.

$(2x+1)^2x^3$	a product	**product**, then chain on $(2x+1)^2$
$\dfrac{\sin x}{x^2+1}$	a quotient	**quotient**
$\sin\left(x^2+1\right)$	sine of something	**cos**(inside), then chain
$\left(x^2 \sin x\right)^4$	a fourth power	**power**, then chain, then product inside

The chain rule is not a competitor to the other rules — it is what comes next. The outermost operation picks the rule you run first; chain is the instruction to then multiply by the derivative of the inside.

WATCH OUT

✗ $\cancel{\frac{d}{dx}\left(\sin x^2\right) = \cos x^2}$ → **the inner derivative is missing — it is** $\cos\left(x^2\right) \cdot 2x$

✗ $\cancel{\frac{d}{dx}(\cos x) = \sin x}$ → **co-functions carry the minus:** $-\sin x$

✗ $\cancel{\frac{d}{dx}\left(\sin^2 x\right) = 2\sin x}$ → **the outer function is *squaring*:** $2\sin x \cos x$

✗ $\frac{dy}{dx}$ ~~is one symbol~~ → dx **and** dy **are differentials — each means something on its own**

✗ ~~the chain rule multiplies by the derivative of the inside~~ → **it is f' *evaluated at* the inside, times the derivative of the inside**

✗ ~~a composite needs one rule~~ → $(2x+1)^2x^3$ **needs product first, then chain**

✗ $\cancel{\left(x^2 \sin x\right)^4}$ ~~starts with the chain rule~~ → **it starts with the *power* rule — chain is what comes next**